# Joseph Wardle - Pipeline Technical Director

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## Work Experience

**Pipeline Technical Director - Student Short Films** October 2025 - Present
Brigham Young University Center for Animation *Provo, UT*

- Develop **Python/Qt artist tools** and a **USD production pipeline** supporting **50+ artists** on Linux and Windows; manage asset and shot data and troubleshoot publishing and DCC workflow failures.

**Sandwich Kwon Do**

- Implemented **immutable, versioned USD publishes** and pinned layer versions at **Tractor** submission, keeping render dependencies consistent and eliminating flicker from assets republished mid-render.
- Built a **Maya previs-to-layout tool** that lets artists review shot changes and validation errors before updating **ShotGrid** and saving versioned layout scenes.
- Built **one-click asset publishing in Maya** backed by a headless Houdini USD builder shared with Substance Painter.
- Developed a shared **Maya/Houdini playblast tool** with FFmpeg burn-ins and **ShotGrid Version uploads**.
- Established consistent **OCIO color workflows** across Houdini, Maya, Nuke, and Blender, adapting OpenDRT and ACES 2.0 transforms for OCIO 2.3 compatibility.

**Honey Business**

- Built **VDB-to-mesh foliage proxies** for entire collections, using USD purposes to switch representations and improve viewport performance by **up to 70×**.

**Web Developer** March 2025 - November 2025
Brigham Young University - Computer Science Department *Provo, UT*

- Developed and maintained **Django web services, REST APIs, and administrative interfaces** for nontechnical users.
- Built **GitHub Actions** workflows for testing, container builds, and deployment; used **Docker Compose** for local development and **Jira** for project tracking.

## Projects

**Cenote - GPU Path Tracer and Hydra Render Delegate** github.com/joseph-wardle/cenote

- Built a **Rust/Vulkan** GPU path tracer and **Hydra 2 render delegate** integrating with usdview, husk, and Houdini.
- Implemented **OpenPBR** shading in Slang; validated energy conservation with **white furnace tests** and rendering results against **pbrt-v4**.

**Javelin - Rigid Body Physics Engine** (github.com/joseph-wardle/javelin) 2023 - Present

- Built a **C++23** physics engine using named modules and a dedicated simulation thread, with an **ImGui/OpenGL** viewer and **Tracy** profiling.

**Film Grain Synthesis** (github.com/joseph-wardle/film_grain) August 2025

- Reimplemented a published film-grain algorithm in **Rust**, with **Rayon** CPU parallelism and **wgpu** compute; benchmarked the CPU implementation across ~7,200 configurations, measuring a **median 72× speedup** over the reference C implementation.

## Education

**Brigham Young University** Provo, UT
*Bachelor of Science, Computer Science (Animation & Games Emphasis)* *August 2023 - Expected May 2027*

- **GPA: 3.95.** Relevant coursework: Data Structures, Multithreading, Linear Algebra, Modeling, Rigging, Shading, FX.

## Skills

- **Programming:** Python, C++23, Rust, MEL, Bash
- **Interfaces & APIs:** Qt/PySide2, Django, REST APIs, ImGui
- **Production Pipeline:** USD, ShotGrid, Pixar Tractor, OCIO
- **DCCs:** Maya, Houdini, Nuke, Substance Painter, Blender
- **Development:** Linux, Windows, Git, Perforce, GitHub Actions, Docker, Jira